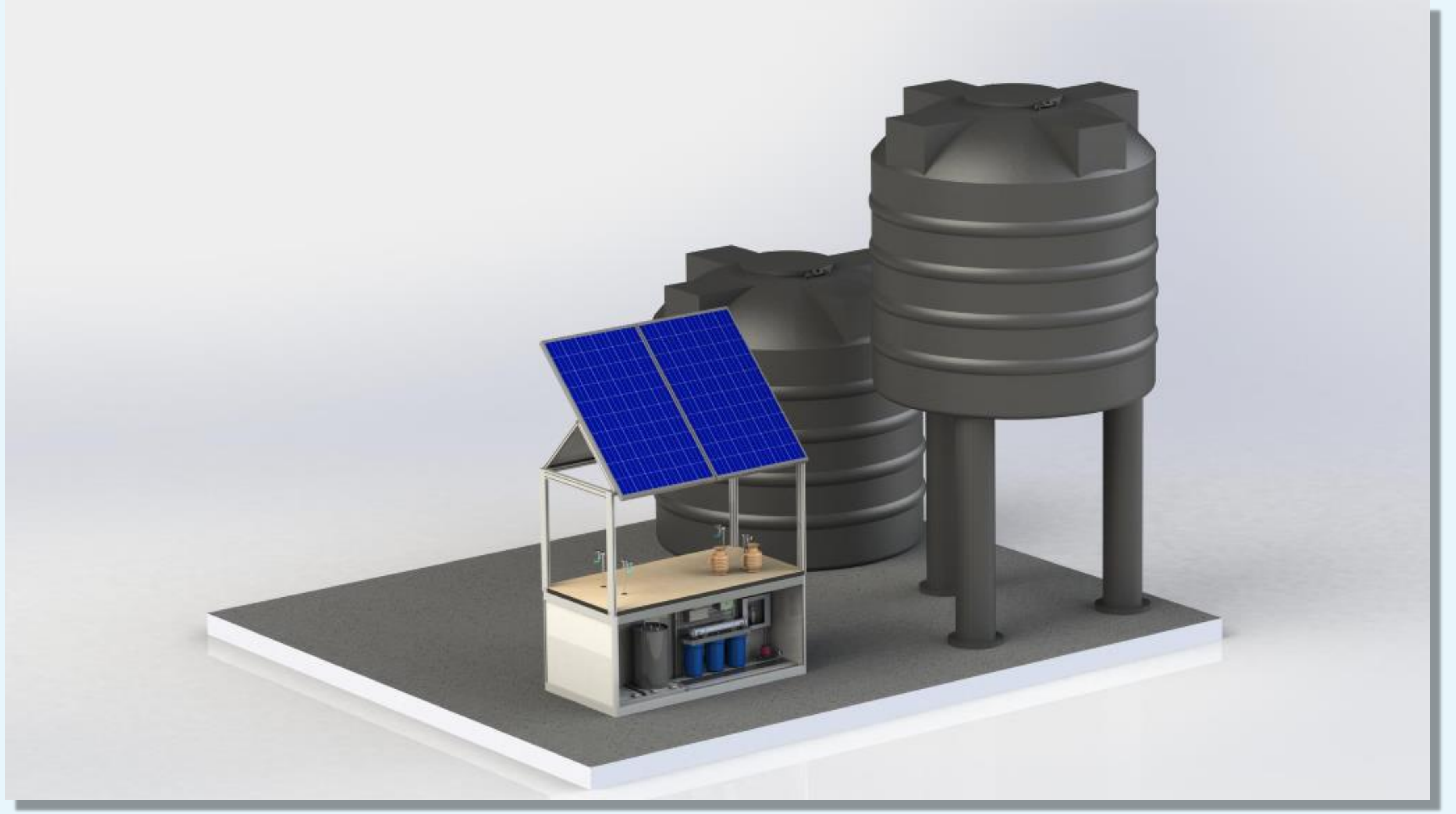
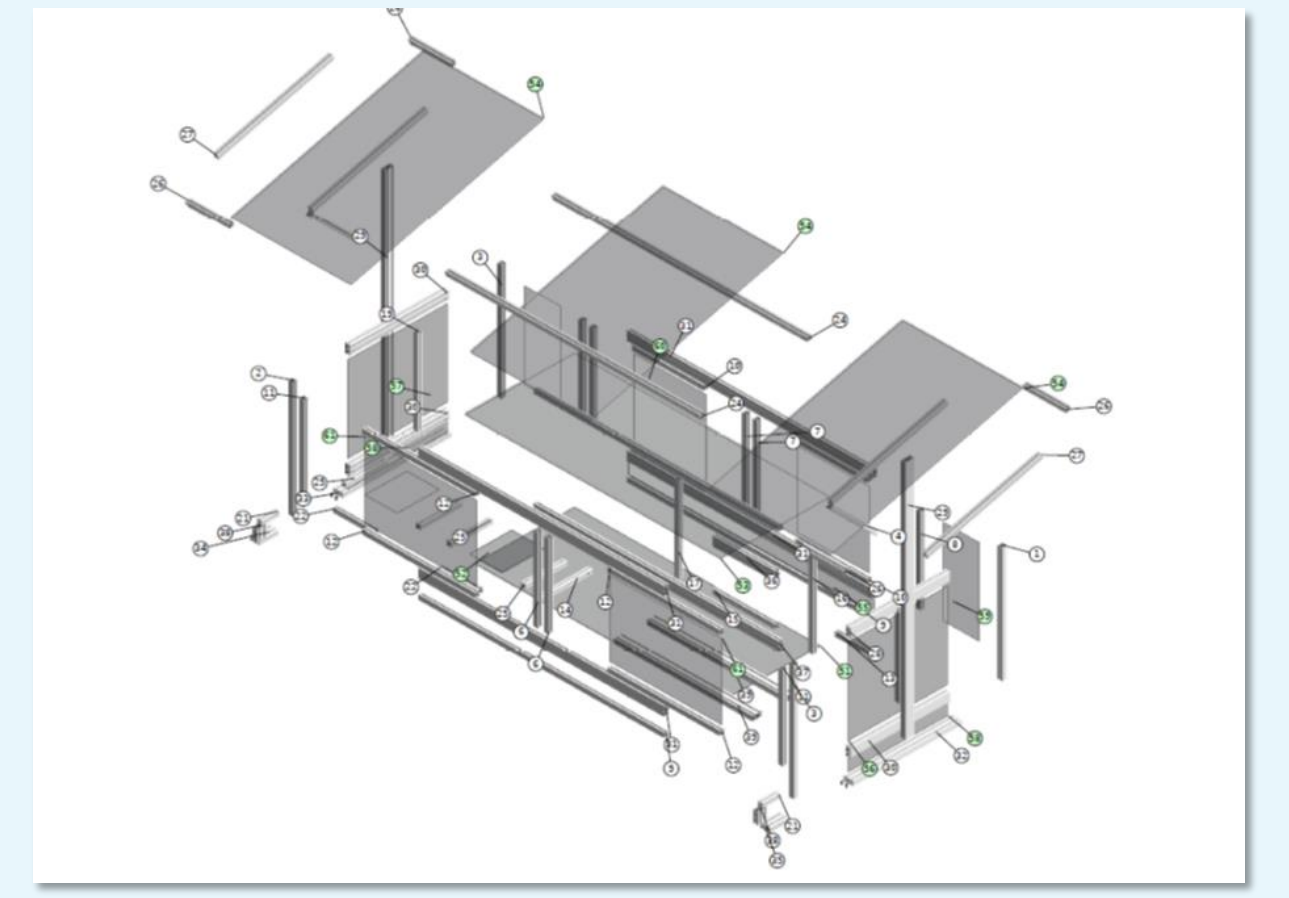
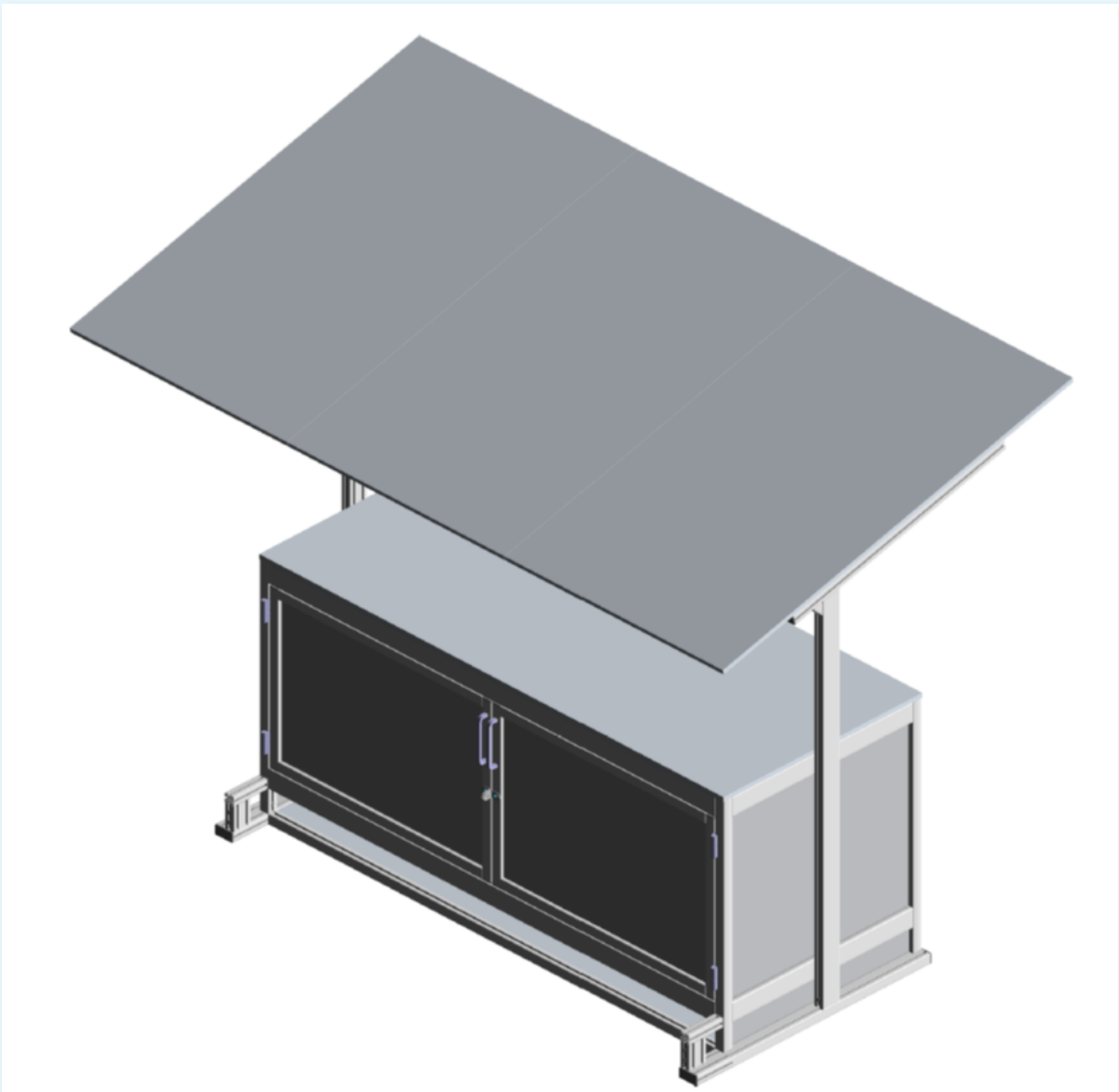
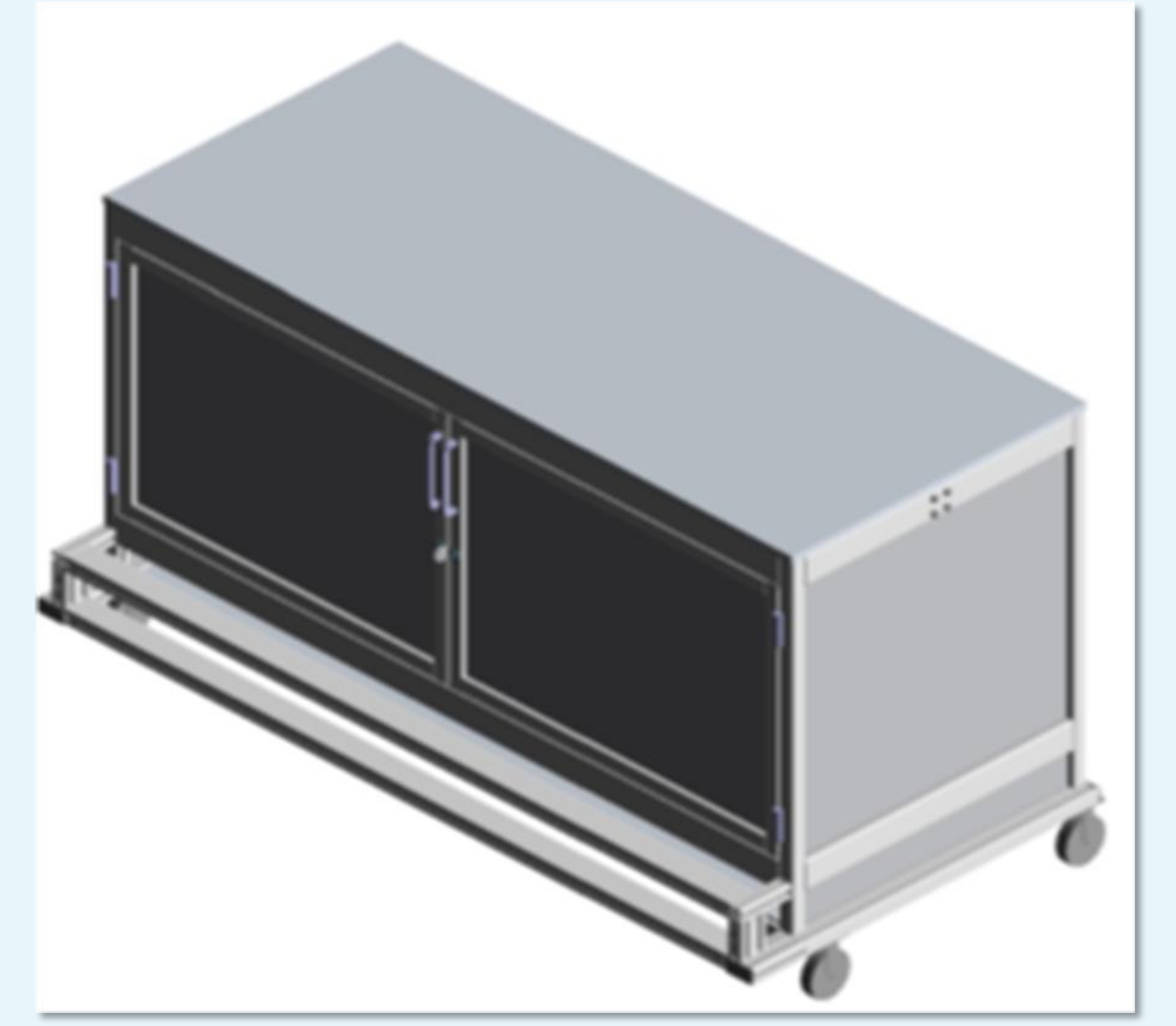
mincarb

Solar Desalination System









Document Number	
Document Name	Solar Desalination Sys v0.5D PFD

Date	Version	Description of Change	c/o #	Author
20/04/2017	v0.5C	Latest Draft		PK
26/11/2019	v0.5D	Generic Variant		PK

Supplier   Manufacturing   Product   Client   Customer

Legend

■	Incoming Inspectiion	■	Metrology/ Test
●	Destructive Test	▼	Temporary Recyclable Material
●	Production Process	▲	Incoming Material
●	Assembly Process	⬚	Enclosed Steps are One Process
■	KPI	■	Outgoing Inspection
▱	Data	■	Packaging
▬	Installation Process		

Flex (0100)

ABS 0100-0001
Flex 0100-0002
Maxim 0100-0003
Digikey 0100-0004
Atmel AVR 0100-0005
Inspection 0100-0100
PCB Flex Assembly 0100-0200
Programming 0100-0300
ETEST 0100-0400
Packaging 0100-0500

Stack Assembly (0200)

Graphite 0200-0001
AC 0200-0002
PTFE 0200-0003
EtOH 0200-0004
IEX-Cat 0200-0005
IEX-An 0200-0006
Mesh Spacer 0200-0007
ABS 0200-0008
XLR Panel Mount 0200-0009
Fasteners 0200-0010
316 SS Plate 0200-0011
Compression Fittings 0200-0012
Potting Compound 0200-0013
Inspection 0200-0100
Paste Production 0200-0200
Lamination 0200-0300
AC Sheet Curing 0200-0400
AC Sheet Curing 0200-0500
Pre-soak 0200-0600
Pre soak 0200-0601
Stack Assembly 0200-0700
Enclosure 3D Printed 0200-0800
Inspection 0200-0850
Install Flex Circuit 0200-0900
Install Interconnects 0200-1000
Install Stack 0200-1100
Install Fittings to Plate 0200-1200
Assemble Plate and Fasten 0200-2000
Leak Test 0200-2100
ETEST 0200-1500
Potting 0200-1400
ETEST 0200-1300

Module Assembly (0300)

Fittings 0300-0001
Copper Bus 0300-0002
Casing 0300-0003
Lexan Sheet 0300-0004
Install bus/ fittings 0300-0100
Cutting 0300-0200
Rack Assembly 0300-0300
Rack Install 0300-0400
Stack Install 0300-0500
System Test 0300-0600

Dynamic Energy Management System (0800)

Power Supply 5V 350A 0800-0001
Prefab Enclosure 0800-0002
Wiring SubAssy 0800-0003
Rear Panel SubAssy 0800-0004
TI DSP 0800-0005
Mincarb Main Board 0800-0006
High Current Relays 0800-0007
Power Resistor 0800-0008
Enclosure 0800-0009
Fan 0800-0010
Copper Bus 0800-0011
Programming 0800-0250
DEMS Barebone Enclosure 0800-0200
DEMS Mainboard 0800-0300
Enclosed Power Resistor 0800-0500
Power Supply Brackets 0800-0020
DEMS Unit Assembly 0800-0400
Power Supply Install Bundle 0800-0150
DEMS Pre-Install Kit 0800-0600
System Test 0800-1100

Fluoride Online Sensor (0900)

Enclosure 0900-0001
4x4 Membrane Button Panel 0900-0002
LCD Display 0900-0003
Trolase Composite 0900-0004
Acrylic 0900-0005
Panel Mount Components 0900-0006
Model Material 0900-0007
Pump Components 0900-0008
Tubing 0900-0009
Fittings 0900-0010
2-Way Valve 0900-0011
Mini Peristaltic Pump 0900-0012
Enclosure Modification 0900-0100
Laser Engrave/Cut 0900-0400
Laser Engrave/Cut 0900-0600
Mount Panel Components 0900-0700
Micro Dosing Pump 3D Printed 0900-2000
Passive Fluidic Circuit 0900-2100
Mount Membrane 0900-0200
Mount LCD 0900-0300
Mount Front Panel 0900-0500
Mount Side Panel 0900-0800
Assembled Dosing Pump 0900-2200
PCB 0900-0013
ATMega2052 0900-0014
Assembled Fluidics 0900-2300
Wiring SubAssy 0900-0015
PCB Fabrication 0900-0900
Programming 0900-1100
Leak Test 0900-2400
PCB Assembly 0900-1000
Controller Install 0900-1200
Mainboard Install 0900-1500
Fluidics Install 0900-2500
Wiring 0900-2600
Etest 0900-3000

Enclosure (0400)

Maytec Alu Profile 0400-0001
Fasteners 0400-0010
Aluminum Composite Panel 0400-0015
Melamine Panel 0400-0020
SS Bench 0400-0021
Seals/ Gaskets 0400-0030
Cutting 0400-0100
Assemble Frame 0400-0200
Install Panels 0400-0400
Bottom Panel 0400-0500
Bench Install 0400-0600
Weather Proofing 0400-0700

Final Assembly (0500)

40mm Cable Ducting 0500-0001
Endurogel Battery 0500-0002
20" Filter 0500-0003
UV Lamp 0500-0004
Main Pump 0500-0005
Solenoid Valves 0500-0006
NFC Reader SubAssy 0500-0007
Wiring SubAssy 0500-0008
HMI 0500-0020
GSM Equipment 0500-0050
Cutting 0500-0150
Incoming Inspection 0500-0100
Install Cable Ducting 0500-0200
Panel Modify 0500-0300
Install PV Management System 0500-0400
Install Battery 0500-0500
Install Pipe Guides 0500-0600
Install Electrode Module 0500-0700
Install DEMS + Electricals 0500-0800
Install Filter Housing 0500-0900
Install UV Lamp 0500-1000
Install Pump 0500-1100
Mount Valves 0500-1200
Install F Sensor 0500-1300
Install Fluidics 0500-1400
Dispenser Installation 0500-1500
Final Wiring 0500-2000
Leak Test 0500-2100
Install HMI 0500-2200
Install COMMS/ Networking 0500-2300
System Test 0500-3000

Packaging/ Freight (0600)

PV Panels 0600-0001
Alu Profile 0600-0010
Product Kit 0600-0030
Shipping Disassembly 0600-0100
Panel Stow 0600-0300
Panel Encap 0600-0400
Final Package 0600-1000
Packaging 0600-0200
Register 0600-2000
Outgoing Inspection 0600-2100
Freight 0600-3000

Site Installation (0700)

Site Preparation 0700-0050
Batteries,PV 0700-0010
Adjustable Angle
Incoming Inspection 0700-0040
Installation 0700-0100
Register 0700-0200
Calibration 0700-0300
First Data 0700-0400

Asset Management (1500)

Filters 1500-0020
Electrodes 1500-0030
Batteries 1500-0010
Reagent 1500-0040
Maintanence 1500-0100

Photovoltaics (1200)

Substrate 1200-0001
40mm Cable Ducting 1200-0002
Hardware 1200-0010
Wiring SubAssy 1200-0050
Cable Duct Install 1200-0100
Hardware Installation 1200-0200
Wiring 1200-0300

Electrical Enclosure (1100)

B&R 1100-0001
Cable Duct 1100-0002
DIN Rail 1100-0003
Panel Mount Items 1100-0004
MeanWell PS 1100-0005
DIN Rail 1100-0006
Transmitters 1100-0007
Wiring Sub-Assy 1100-0008
Cutting and Prep 1100-0100
Cable Ducting 1100-0200
DIN Rail Install 1100-0300
Cable Gland And Switches 1100-0400
Mains Supply 1100-0500
DIN Components 1100-0600
Sensor Transmitters 1100-0700
Wiring 1100-0800

Main Fluidics (1300)

Sch80 DN 20 GF++ 1300-0001
Sch80 Fittings GF++ 1300-0010
Sensors 1300-0020
Dosing Pump 1300-0030
Pipe Cutting 1300-0100
Pipe Subassembly 1300-0200
Sensor Installation 1300-0300
Dosing Pump Installation 1300-0400

Dispenser (1400)

Sch80 Fittings GF++ 1400-0003
Flowmeter/Valve 1400-0002
Sch80 Piping GF++ 1400-0001
Dispenser Sub Assembly 1400-0100